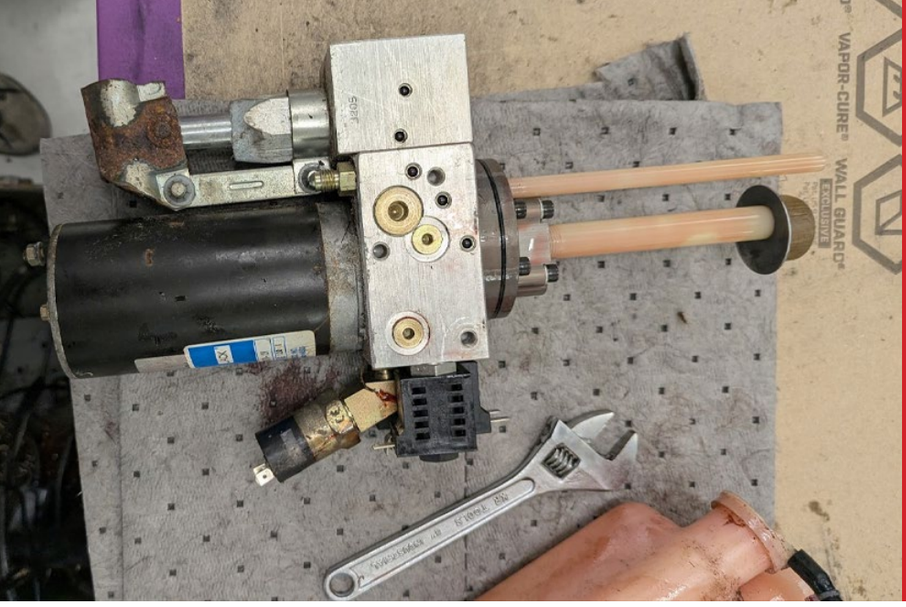

FLUID MACHINE DISSECTION

FINAL PROJECT

Summary

For my Intro to Fluid Dynamics class, we were given a final project called a fluid machine dissection. For this project, the students would be grouped and tasked with disassembling and highlighting the fluid mechanical concepts we found relevant to our coursework. My team was given a Monarch Dyna-Jack Hydraulic Power Unit M-268-0108 30170-12V, which we took time in class to tear apart and analyze. In mechanical terms, this machine was used to power hydraulic rams in wheelchair lifts during World War I. It would either be powered by electricity or a mechanical lever, to move a fluid called Mobil DTE 24 through a gear pump and into a hydraulic jack. The hydraulic ram would move due to an influx of fluid on one side of a bearing, which caused a difference in pressure that made the bearing move to extend or contract.

In terms of fluid dynamics, we analyzed the streamline flow through the tubes and gear pump. We noted that fluid through the tube moved as a Poiseuille flow, making assumptions such as mean flow to disregard the no-slip condition of liquids. We also made the assumption that the hydraulic lines were non-deforming with pressure and that they had a constant cross-sectional area. With these assumptions, we were able to then calculate the mechanical advantage for the hydraulic rams with just a function of pressure and the relative areas within the ram. We determined that the overall flow through the system was created by a pressure difference created by a gear pump, which would cause the fluid to move from higher pressure regions to lower ones.

If you would like a deeper dive into our analysis, then I have the video attached below:

https://www.youtube.com/watch?v=HcbzdTH58LI